tions by rigorously ensuring that subsystems are unbypassable and tamperproof. For example, type enforcement may be used to implement assured pipelines to provide these properties. An assured pipeline ensures that data flowing from a designated source to a designated destination must pass through a security-related subsystem and ensures the integrity of the subsystem. Many of the security requirements of these applications may be ensured by the underlying mandatory security mechanisms of the operating system.

Operating system mandatory security mechanisms may also be used to rigorously confine an application to a unique security domain that is strongly separated from other domains in the system. Applications may still misbehave, but the resulting damage can now be restricted to within a single security domain. This confinement property is critical to controlling data flows in support of a system security policy. In addition to supporting the safe execution of untrustworthy software, confinement may support functional requirements, such as an isolated testing environment in an insulated development environment.

Although one could attempt to enforce a mandatory security policy through discretionary security mechanisms, such mechanisms can not defend against careless or malicious users. Since discretionary security mechanisms place the burden for security on the individual users, carelessness by any one user at any point in time may lead to a violation of the mandatory policy. In contrast, mandatory security mechanisms limit the burden to the system security policy administrator. With only discretionary mechanisms, a malicious user with access to sensitive data and applications may directly release sensitive information in violation of the mandatory policy. Although that same user may also be able to leak sensitive information in ways that do not involve the computing system, the ability to leak the information through the computing system may increase the bandwidth of the leak and may decrease its traceability. In contrast, with mandatory security mechanisms, he may only leak sensitive information through covert channels, which limits the bandwidth and increases accountability, if covert channels are audited.